# WorkProof: Project Documentation

**Proof you can do the job, not just a CV.** Credify System Thinking Challenge 2026 · Area 04: Graduate employability

Oct 28, 2026

## Summary

WorkProof helps employers spot capable graduates by testing the work itself, not the CV. An employer pastes a job ad they already have. WorkProof turns it into a 45-minute work sample that candidates take on their phones, scored blind against a written scoring sheet, with a receipt for every decision.

The argument in one line: employers don't lack information about graduates, they are drowning in it. So instead of asking graduates for more evidence, WorkProof makes **being wrong about a graduate cheap**.

It is built and running: a live app at workproof-gray.vercel.app, a public repository, a live database, and 19 automated tests. This document covers the ten topics the challenge asks for, in order.

## 1. Problem and context

Graduate unemployment in South Africa rose from 10.3% to 12.4% in two quarters, even though hiring young people is already subsidised.

| Measure (Stats SA QLFS) | Q2 2026 |
|---|---|
| Graduate unemployment | 12.4% (10.3% in Q4 2025, 12.2% in Q1 2026) |
| Official unemployment | 33.6% |
| Youth unemployment, ages 15 to 34 | 47.4% |
| Youth unemployment, ages 15 to 24 | 62.8% |
| Not in employment, education or training, ages 15 to 24 | 36.4% |

The challenge question is how employers can identify capable graduates who lack traditional work experience. Most junior job ads answer it with requirements that screen out exactly those graduates: years of experience, degree class, school marks, own

transport.

**The experiment South Africa already ran.** The Employment Tax Incentive (ETI) cuts the PAYE an employer pays for workers aged 18 to 29, claimed monthly on the EMP201. It is live in 2026. So the state has already made it cheaper to take a chance on a young person, and graduate unemployment went up anyway. Cost was never the whole constraint. The employer still cannot tell who is capable, so a junior hire feels risky whatever it costs.

We make no claim about how many employers use the ETI; we found no uptake data. The claim is only that the incentive exists and the problem persists.

## 2. Stakeholders

Only one stakeholder makes the hiring decision, the hiring manager, and they also carry most of the risk. Everyone else works around that decision.

| Stakeholder | Wants | Fears or constraint | Power over the hire |
|---|---|---|---|
| Hiring manager (often the owner, in a small business) | Someone who can do the work in month one | A bad junior hire costs months of supervision, in public | Decides |
| Graduate | A first chance to show what they can do | Filtered out before anyone sees their work | None: applies, then waits |
| Recruiter or HR | A short shortlist, fast | Too many applications to read properly | Filters, usually on years and degrees |
| Government (ETI, SARS) | Young people in work | A subsidy that doesn't move the number | Lowers the cost of hiring, not the risk |
| Evidence platforms (universities, portfolio sites, Credify) | Graduates placed | More evidence that nobody has time to read | Inform the decision, but don't make it |

The hiring manager's incentives explain the rest. A bad hire is visible and costly. A good graduate they turned away costs them nothing anyone will ever measure. Filtering hard on experience is therefore rational behaviour, not laziness or prejudice.

Most interventions target the graduate. WorkProof targets the hiring manager, because that is where the decision and the risk sit.

## 3. Root causes

There are two root causes, and both sit with the employer, not the graduate: too much evidence to read, and a risk that only runs one way.

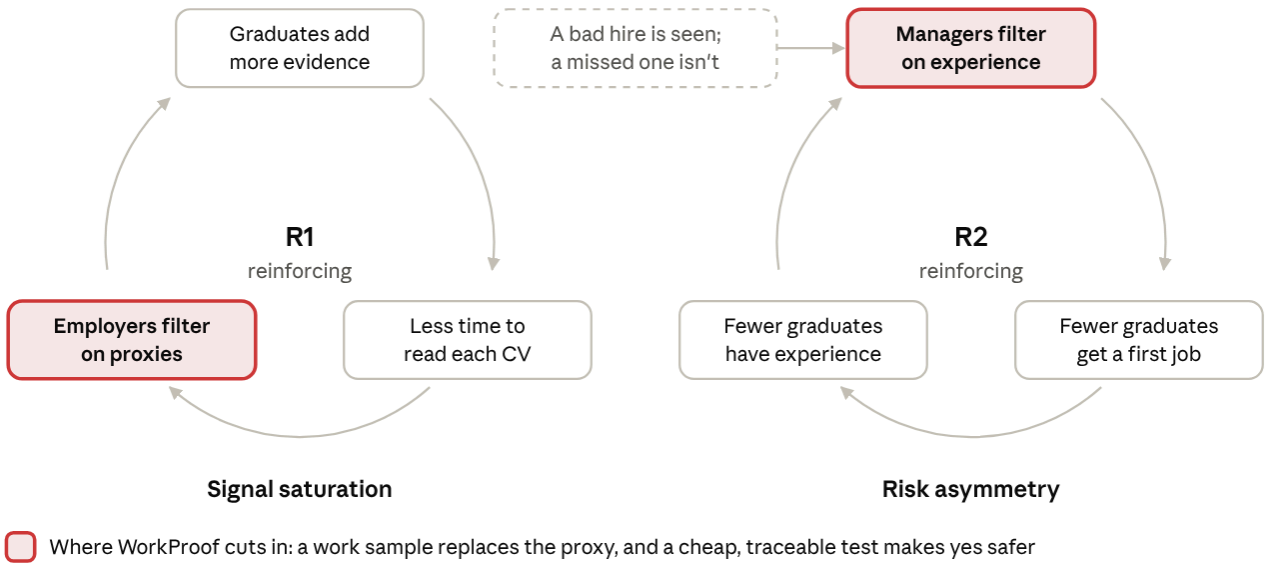
**What we mean by a proxy.** A proxy is a requirement that stands in for ability without measuring it. "Minimum 2 years' experience" is a proxy: it hopes the person can do the job because they did something like it before. "Reconcile the bank statement monthly" is not a proxy; it is the job. Common proxies in junior ads: years of experience, degree or degree class, school marks ("Matric with 70%+"), and circumstance ("own reliable transport"). Proxies are cheap to check, which is why they are used. They also screen out every capable graduate who hasn't had a first job yet.

**Root cause 1: signal saturation.** A junior role can draw more applications than anyone can read with care (application volumes were not measured; see Limitations). Each new evidence channel (portfolios, certificates, badges) makes the pile heavier, not clearer. With seconds per CV, employers fall back on proxies, and graduates respond by adding still more evidence. The candidates who most need to stand out have the least ability to buy attention.

**Root cause 2: risk asymmetry.** A bad junior hire costs a manager months of supervision and visible damage to their reputation. A missed good hire costs them nothing measurable. When the costs are that lopsided, over-filtering on degree, experience and referral is the rational choice. Handing the graduate more paperwork cannot fix an incentive like that.

## 4. System relationships

The two root causes form two reinforcing loops, so the problem feeds itself: each turn makes the next round of filtering look more justified.




system map · two reinforcing loops, one outside driver

In R1, more evidence means less time per CV, which pushes employers onto proxies, which pushes graduates to add more evidence. In R2, filtering on experience means fewer graduates get a first job, so fewer have experience, so the filter looks justified. R2 is driven from outside by the risk asymmetry: a bad hire is seen and a missed one isn't.

**Evidence from live job ads (28 October 2026).** The loops show up in the ads themselves. Every job ad on CareerJunction carries a level field that the employer fills in, so a title can be checked against the employer's own rating.

| Label in the job title | Live ads | Ranked above junior by the employer's own level field |
|---|---|---|
| "Junior" | 30 (of 59 results for "junior") | 10, or 1 in 3 (8 Intermediate, 1 Senior, 1 Executive) |
| "Graduate", "trainee" or "intern" | 7 | 1 (14%) |

On Pnet, an ad titled "Junior contracts manager" asked for a "minimum 8–10 years' experience in construction". "Junior" has become the word for experienced but cheap.

**Where the leverage is.** Each actor pushes on a different point of the system. The ETI changes a parameter: the cost of a hire. Evidence platforms add to information flow, a channel already saturated. WorkProof changes the rules of the decision: what is checked (the work, not proxies), who carries the cost of evaluating (a reusable kit instead of reading every CV), and what is recorded (a receipt). In Meadows' terms, rules are a

stronger leverage point than parameters or information. WorkProof also adds a third loop, a balancing one: 90-day check-ins feed back into the kit, so the test improves with every hire.

## 5. Intervention

WorkProof is an employer-side evaluation kit: a job ad goes in, and a short work sample, a written scoring sheet and a decision receipt come out. The tool drafts and organises; a person scores and decides. It never ranks candidates.

1. **Ad X-ray.** Written rules mark each line of the ad as a proxy, trainable (a software tool) or a real task. They flag ads titled "Junior" whose requirements read Intermediate or Senior, and raise a fairness flag on proxies that screen by circumstance. The employer also gets the ad rewritten without proxies, ready to copy. Rules, not AI, so every flag can be explained.
2. **Kit.** The real tasks become up to three abilities, a 45-minute test and a four-level scoring sheet with every level written out. The employer can edit every word before sending; once sent, the kit locks.
3. **Live candidate link.** A link and a QR code. Candidates take the test on their own phones, with no account. In the bookkeeping kit, each candidate's ledger carries three of six possible planted errors (20 combinations), so answers can't be passed around.
4. **Blind scoring.** Answers arrive in about two seconds, shown by candidate number only. The scorer gets an answer key for that candidate's planted errors and one suggested live follow-up question. Names unlock only after every answer is scored.
5. **Decision receipt.** Scores, the answer key, declared AI use, the follow-up answer and the decision, plus written feedback for the candidate built from the scoring sheet. An "invite" decision shows a note that the hire may qualify for the ETI. Candidates leave an email when they start; like their name, it stays hidden until every answer is scored, and then one button opens the employer's email app with the invite or the feedback already written.

Two features close the loop:

- **The filter you didn't use.** Candidates state their years of experience at the end of the test; the scorer never sees it. Once names unlock, WorkProof shows what the ad's experience rule would have done to these real candidates, for example: "Your ad's '2+ years' rule would have rejected your top scorer."
- **90-day check-in.** After a hire, the employer records how it is going. The inbox lists check-ins against test scores, so over time the employer learns whether the test predicts the job, and sharpens it.

**The intervention we rejected: a graduate portfolio platform.** In Donella Meadows' leverage points, a portfolio platform works on information flow, a weak point to push on because the channel is already saturated. WorkProof works on the rules: who carries the cost of evaluating, and how the decision is made. Information platforms, Credify included, are necessary but not sufficient; WorkProof sits on top of them rather than against them.

**How this differs from existing skills-test platforms.** Online assessment sites such as TestGorilla and Vervoe already exist; they mainly offer libraries of ready-made tests. WorkProof works differently in three ways. It starts from the employer's own job ad and removes the proxies in it, so the test comes from the real duties. It shows the employer what their experience rule would have done to real candidates ("the filter you didn't use"), which is the change in judgement this project is about. And it is free and built for small South African employers, linked to the ETI they can already claim.

## 6. Feasibility

WorkProof already runs end to end on free infrastructure; the open question is adoption, not technology.

| Aspect | Status |
|---|---|
| Running cost | R0 on Supabase's free tier |
| Speed | About 2 seconds from a candidate's Submit to the employer's inbox |
| Candidate access | One link or QR code; any phone; no app, no account |
| Test | 45 minutes |
| Quality | 19 automated tests on the X-ray rules and ad rewrite, planted errors and answer key, rules kit, the experience-filter comparison and the result email |
| Security | Row-level security on, no policies; every read and write goes through 6 locked database functions; a per-invite employer key never leaves the employer's browser |
| AI | Optional and free. An open-weight model (openai/gpt-oss-120b) on Groq's free tier drafts kits for new ads, seeing only the ad's title and duties; without a key, written rules draft the kit |

| Aspect | Status |
| --- | --- |
| Stack | React 19, TypeScript, Vite, Tailwind CSS 4, Supabase (Postgres, Realtime, Edge Function) |

**Technical.** The full flow has been tested in the browser against the live database: invite, candidate submit, live inbox, blind scoring and reveal, kit editing, ad rewrite, filter comparison and check-in. Scores are validated on the server and merged per ability, so two quick saves can't overwrite each other.

**Economic.** The cost to an employer is reviewing the kit once, then scoring time per candidate. That replaces reading a pile of CVs, and the kit is reused for every candidate on the shortlist.

**Adoption.** The ETI gives employers a reason to hire young people that already exists. WorkProof lowers the risk side of the same decision, and reminds the employer of the ETI at the moment they choose to invite a candidate.

## 7. Risks and trade-offs

The biggest risks are candidates using AI to answer, bias in scoring and leaked candidate data; each has a design response built into the product.

| Risk | What WorkProof does |
| --- | --- |
| Candidates let AI do the test | AI is allowed and declared, as at work. Each candidate gets a different set of planted errors, and one live follow-up question ("Why did you flag that?") checks they understand what they found. |
| Bias in scoring | Scoring is blind: candidate numbers only until the last score. Age is never asked. Proxies that screen by circumstance get a fairness flag. |
| Candidate data leaks | The browser cannot read any table directly; every read goes through locked functions checked by the invite code or the employer key. The realtime channel carries a "changed" ping, never answer data. |
| The tool becomes the decision | The kit drafts; a person scores against written levels and makes the decision. WorkProof never ranks candidates. |

| Risk | What WorkProof does |
|---|---|
| Spam submissions on a shared link | Not yet mitigated: anyone with the link can submit. A real deployment needs rate limiting. |
| Small numbers mislead | Check-ins are shown as a signal, not a statistic. |

**The main trade-off: candidate time.** A 45-minute test asks more of a candidate than uploading a CV. In return, candidates skip the CV and experience filter entirely, and every one of them gets written feedback, which most applicants never receive. If employers sent the test to hundreds of applicants, that time would add up; a sensible rollout sends it to a shortlist chosen without proxies.

**Fairness trade-off: phone access.** Taking the test needs a phone with data for 45 minutes. That is a lower bar than own transport, but it is still a bar, and should be watched in a pilot.

## Security, misuse and privacy

WorkProof is built so that the people who could misuse it gain little, and so that it collects as little as possible.

| Misuse or threat | What WorkProof does | Still open |
|---|---|---|
| Employer uses the "test" to get free work | Tests use made-up data (a fictional shop's ledger, invented scenarios), capped at 45 minutes, so the answers are worthless as real work. | An AI-drafted task is reviewed by the employer, not by WorkProof. |
| Fake employer posts a job to harvest candidate details | Only a name and email are collected, and they stay hidden until every answer is scored. No ID number, CV or address. | Employer verification is a next step. |
| Someone hides instructions in a job ad to manipulate the AI | Only the ad's title and duties are sent, capped in length; the AI must answer in a fixed format that is checked; the employer reviews and can edit every word. | |
| Candidates share answers | Each candidate gets a different set of planted errors (20 combinations). | Only in the bookkeeping kit so far. |

| Misuse or threat | What WorkProof does | Still open |
|---|---|---|
| Someone else takes the test for a candidate | One live follow-up question in the interview checks the candidate understands their own answers. | |
| Candidates use AI | Allowed and declared, as at work; the planted errors and live question test understanding. | |
| Spam submissions on a shared link | Submissions are validated on the server. | No rate limiting yet; it is a next step. |
| Someone reads another employer's data | Tables have no open access; every read goes through locked functions that check the link code or the per-invite employer key. The live channel carries no data. | No employer accounts yet; the key lives in one browser. |
| Stolen API key | The Groq key is stored only as a server secret; the browser never sees it. The public database key grants nothing on its own. | |
| Injected code in emails | Everything a person types is escaped before it goes into the designed email (tested). | |

**What is captured and stored.** Candidates: name, email, answers, AI tools used, years of experience, time taken. Employers: the job ad, the kit, scores, decisions and check-ins. Stored in a Supabase database in the EU (Ireland). The employer key and a candidate's test progress stay in their own browser. Never collected: ID number, age, race, gender, address, photo, CV, school results, cookies or trackers. Candidate answers and contact details are never sent to an AI service. The app's privacy notice says this in plain words: workproof-gray.vercel.app/privacy. It is designed around POPIA's principle of collecting only what is needed; automatic deletion after a set period is not built yet.

## 8. Assumptions

WorkProof rests on five assumptions; each comes with the finding that would weaken it.

| Assumption | What would prove it wrong |
|---|---|
| Junior job ads often ask for more experience than a junior can have | Ads turn out to be mostly reasonable about experience. Our snapshot says otherwise (1 in 3 "junior" ads ranked above junior), but it is one day on two job boards. |
| Hiring managers over-filter because a bad hire is costly and visible, while a missed good hire is not | Managers report that missed good hires are measured where they work. |
| Employers don't already run structured work samples for junior roles | Employers say they already do. Then the gap is adoption, and the product should target that instead of design. |
| A short work sample predicts junior performance better than years of experience or an unstructured interview | 90-day check-ins show no link between test scores and how hires do. WorkProof measures this itself. |
| Employers will spend time scoring if it replaces CV screening | Employers abandon kits after sending them. A pilot will show this quickly. |

The last two are testable inside the product, which is why the 90-day check-in exists.

## 9. Limitations

WorkProof is a working prototype, not a tested product: it has no employer accounts, one role with planted errors, and no real hires yet.

**Product**

- No employer accounts. The employer key lives in one browser; clearing site data loses access to that link's answers.
- Anyone with a candidate link can submit, as with a shared form. A real deployment needs rate limiting.
- Planted-error tests exist only for the bookkeeping sample. Other roles get a written task scored against the sheet, with no automatic answer key.
- The answer key is an aid, not a grade. It matches on rand amounts and on what each error is about; the scorer decides.
- The X-ray reads bullet points under "Requirements" and "Duties" headings. Ads written as prose need reformatting first.
- Years of experience are self-reported, and used only for the comparison after scoring.
- Two candidates starting in the same second could draw the same candidate number.

- The database is in Ireland (eu-west-1), the closest free region, so live updates take about two seconds from South Africa.

**Evidence**

- The job-ad data is one day's snapshot (28 October 2026) from two job boards, with recruiter-posted ads over-represented.
- The "graduate, trainee or intern" comparison rests on 7 ads: a pattern, not a rate. No data on how many applications a junior ad receives was collected, so the saturation loop (R1) rests on reasoning, not measured volumes.
- The Pnet count is a floor: 8 of 24 junior ads read state an experience requirement, but 16 could not be checked.
- No employer or graduate has used WorkProof in a real hire yet, so its effect on hiring is untested.

## 10. Expected impact and measurement

The expected impact is more graduates with no experience reaching a shortlist and getting hired, measured inside the product from the first test.

| What we measure | How | In the app today? |
|---|---|---|
| Graduates shortlisted with 0 years' experience | The filter comparison counts candidates the ad's rule would have rejected, and how they scored | Yes |
| Whether the test predicts the job | 90-day check-ins compared with test scores | Yes |
| Ads rewritten without proxies | Copies of the rewritten ad | Not yet: the copy button exists, the counter does not |
| Feedback reaching candidates | Receipts with feedback sent after each decision | Not yet: feedback is generated, sending is not counted |

**Who benefits.** Graduates get judged on their work and receive specific feedback even when they don't get the job. Employers find capable people their ad would have filtered out, and get a record of every decision. Government gets more from an incentive it already pays for.

**The feedback loop.** Every 90-day check-in shows whether the test predicted the job. An employer whose high scorers struggle can change the kit, so the test gets better with each hire. This is the balancing loop WorkProof adds to the system: it pulls the test toward what actually predicts the job.

**Next steps**

1. Pilot with small employers hiring junior bookkeepers.
2. Add employer accounts, rate limiting, and the two missing counters.
3. Build planted-error kits for more entry-level roles.
4. With the candidate's consent, share verified results with evidence platforms such as Credify.

## Key decisions and why

Each major design choice, the alternative considered, and the reason for the choice.

| Decision | Alternative considered | Why |
|---|---|---|
| Build for the employer | A tool to help graduates show their skills | The employer decides and carries the risk; graduates already do everything they are told. |
| Reject the portfolio platform | Graduates upload projects and badges | It adds evidence to a channel nobody has time to read. Information is necessary, not sufficient. |
| Test the real work | Better CV screening or interviews | A work sample shows ability directly and predicts the job better than years of experience. |
| Rules, not AI, for the X-ray | AI judging the ad | Every flag can be explained, and it works with no AI at all. |
| Blind scoring | Seeing names while scoring | Names, emails and experience stay hidden until the last score, so they cannot sway it. |
| Allow AI in the test | Ban AI and try to catch it | People use AI at work. Unique planted errors and a live question test understanding instead. |

| Decision | Alternative considered | Why |
|---|---|---|
| Show "the filter you didn't use" | Simply hiding the experience rule | Employers see, in their own data, who the old rule would have rejected. That is what changes judgement. |
| No logins | Candidate and employer accounts | Any barrier cuts out the graduates this is for. Employer accounts are a next step. |
| Groq's free AI for new kits | A paid model per kit | Free keeps it workable for small employers; rules take over if AI is unavailable. |
| Template result emails | AI-written emails | Feedback to candidates should only say what the scores show. |
| Hosted live on Vercel | Running only on a laptop | Reviewers can try it with nothing to install, and the QR code works on any phone. |

## Sources and how it was built

**Repository:** github.com/yaseenselemane/workproof: code, README, tests, database migrations, and the job-ad data as CSV files in `docs/`.

**Sources**

- Statistics South Africa, Quarterly Labour Force Survey, Q2 2026 (released 11 August 2026). The 12.4% graduate figure is in the full statistical release (P0211, section 4.1); the Q4 2025 figure of 10.3% is from the Q4 2025 presentation.
- Employment Tax Incentive: SARS. SARS employer guide PAYE-GEN-01-G05: for employees aged 18 to 29, up to R1,500 a month off the employer's PAYE in the first 12 months (rates from 1 April 2025); the incentive runs until 28 February 2029.
- Job ads: every live CareerJunction result for "junior" on 28 October 2026 (59 ads; 30 with "Junior" in the title), plus 24 junior ads read on Pnet the same day.
- Donella Meadows, *Leverage Points: Places to Intervene in a System* (1999).

**How it was built.** Yaseen Selemane chose the challenge area and the angle, challenged the research method, set the product bar that every feature must actually work, chose the visual direction, and decided scope and priorities. WorkProof was built by him, with Claude Code (Anthropic's AI coding assistant) as his main engineering tool. Working to

those decisions, Claude Code drafted the research and analysis, wrote the application code and tests, and drafted the documentation. The full disclosure is in `AI_USE.md` in the repository.